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Reflections on Data Science 2026

Journal

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#1. Lead author: Collective, Calling BS on "People who eat a late dinner may gain weight".

(What is the claim?)

The article claims that eating dinner late leads to higher blood sugar levels and less fat burning, suggesting that late eating may contribute to obesity and diabetes [1]. It presents meal timing as an important factor in metabolic health and implies that eating late is inherently harmful.

(Why is it BS?)

The article is an instance of a lack of alternative hypotheses, combined with some statistical traps, interpreting short-term experimental results as evidence for long-term health effects.

The study is based on only 20 healthy participants, which is too few to support conclusions about population-level health effects. Individual metabolic differences can be large, and a sample size of 20 cannot capture them. However, the main problem is not only the sample size, but also the experimental setup. Participants ate at 22:00 and went to sleep at 23:00, meaning the "late eating" condition is confounded with eating immediately before sleep [2]. This design makes it unclear whether the observed effects are due to meal timing, closeness to sleep time, or circadian factors. Furthermore, the study measures only short-term metabolic responses to a single meal, yet the article discusses long-term outcomes such as obesity and diabetes. This is a reach; acute metabolic changes do not necessarily translate into chronic health effects without real evidence. Although the authors acknowledge that more research is needed to determine whether these effects persist, this uncertainty is not reflected in the article's framing. In addition, the reported differences are presented without context regarding variability or clinical significance. Without information about distribution or effect size relevance, the results may appear more meaningful than they actually are. Finally, the participants were all healthy individuals, yet the article speculates about stronger effects in people with obesity or diabetes. This extends the conclusions beyond the studied population and introduces unsupported assumptions.

(Which tools have you used?)

Several tools were applied to evaluate the article. The tool of generating alternative hypotheses was applied by considering whether the observed metabolic differences could be explained by circadian rhythm effects or by eating close to bedtime rather than by meal timing itself as the primary causal factor. We also identified generalization, where results from a small, controlled experiment are extended to a general population and long-term health outcomes. Lastly, we used statistical reasoning to identify the presence of statistical traps. Particularly, the absence of variability reporting and unclear significance of the reported effects can make findings seem stronger than they actually are.

(What are the (un)intended consequences of spreading this BS/misleading information?)

Presenting late eating as inherently harmful may lead to public-health messages that overemphasize meal timing while underestimating more important factors such as total calorie intake, diet quality, and overall lifestyle. It may also create unnecessary anxiety around meal timing, encouraging restrictive or disordered eating behaviors. Altogether, this can distort how people understand complex metabolic processes, leading them to believe that small timing differences have significant health consequences.

References

- [1] Endocrine Society, “People who eat a late dinner may gain weight,” Press Release, Washington, DC, Jun. 11, 2020. Available: <https://www.endocrine.org/news-and-advocacy/news-room/2020/people-who-eat-a-late-dinner-may-gain-weight>
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#2. Lead author: Lucie Navratilova, Calling BS on "Microsoft's AI Is Better Than Doctors at Diagnosing Disease".

(What is the claim?)

The article reports that Microsoft's AI diagnostic system MAI-DxO correctly diagnosed 85% of complex medical cases, compared to 20% for human doctors. This implies that AI dramatically outperforms doctors and could soon transform healthcare by delivering highly accurate, cheaper, and faster diagnoses.

(Why is it BS?)

However, the doctors were explicitly instructed not to use external resources such as search engines, textbooks, colleagues, or language models. Additionally, the cases used were picked from the New England Journal of Medicine's diagnostic challenges, which are intentionally designed to be rare and difficult cases for the sake of teaching. They are not representative of routine clinical practice. So this experiment is actually testing how well someone can remember things when they're put under strict constraints. It's not really measuring how they would perform in a real diagnostic setting. Secondly, the AI system was not a single model, but a pooled set of LLMs coordinated by an "orchestrator" that simulated consultation among experts. In contrast, the human comparison group consisted of 21 individual doctors working alone. This created an unfair comparison - a collective multi-model system versus individual doctors without a specialist support, which is also unrealistic in practice. Thirdly, the article emphasizes that the AI was "four times more accurate" but it does not establish an appropriate null baseline. It is unclear how specialists or collaborative medical teams would perform on the same cases. Without these comparisons, it is statistically incomplete. Additionally, the reported 20% accuracy of doctors is reported as a single mean value without reporting variance or the distribution, making it unclear whether the performance was uniformly low or highly variable. Finally, the cost comparison was done in a simulated setting, which does not reflect real-world medical practice, including complications, legal risks, or the complexity of actual health care systems.

(Which tools have you used?)

To call bullshit on the article, we applied some tools from the course. We applied the concept of unfair comparison, it is basically an apples-to-oranges comparison: a collective multi-model AI system and individual doctors without specialist input. A fair comparison would require either specialist physicians or medical teams as the control group. Secondly, we looked at the absence of a proper null model or baseline for specialist performance. Thirdly, they don't report any measure of variability. Showing only the average, without indicating how much the data actually vary, can make differences look clearer or more meaningful than they really are. Lastly, we considered whether strong performance on carefully written academic case studies actually translates to real-world medical practice.

(What are the (un)intended consequences of spreading this BS/misleading information?)

Posing AI as dramatically superior to doctors may harm public perception of medical practice and create unrealistic expectations about automated diagnoses, and it may harm the public's trust in general doctors. Overstated claims could influence investment and regulatory decisions. While AI systems may be a valuable addition to the medical field, conclusions about replacing or outperforming doctors are harmful as they oversimplify the complexity of medical care and undermine trust in doctors.

#3. Lead author: Matús Stanko, Calling BS on "American steel is BACK.", The White House on X, (2026).

(What is the claim?)

In January 2026, the official White House account on X posted a bar chart named “Total U.S. Steel Production (Mt) 2024 v. 2025”, announcing in the caption “American steel is BACK” [1]. The chart visually shows that U.S. steel production has increased dramatically, with the 2025 bar appearing several times taller than the 2024 bar.

(Why is it BS?)

This is a case of a misleading visualization caused by a truncated y-axis. Rather than starting at zero, the y-axis begins at 80.2 Mt, which makes the difference between the two years appear much larger. The actual production values are 80.7 Mt in 2024 and 81.8 Mt in 2025 — a real increase of only around 1.1 Mt, or less than 1.4%. Despite the small change, the graph makes the increase appear much larger than it actually is. This is a deliberate design choice that misleads the viewer into believing that there is a dramatic industrial revival, but the data does not support this.

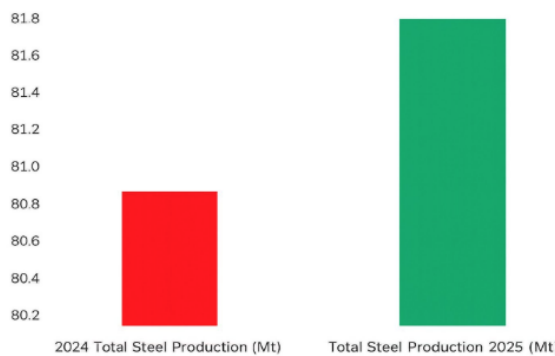
(Which tools have you used?)

I applied Tufte’s *Lie Factor* [2] to quantify the distortion introduced by the truncated y-axis. I also re-drew the visualization with the y-axis starting at zero. The comparison between the original and corrected charts is shown in Figure 1. Using Tufte’s definition, I estimate the distortion as $\text{Lie Factor} \approx \frac{2.2}{0.0136} \approx 161.5$, meaning that the chart makes the increase look around 161 times larger than it really is.

(What are the (un)intended consequences of spreading this BS/misleading information?)

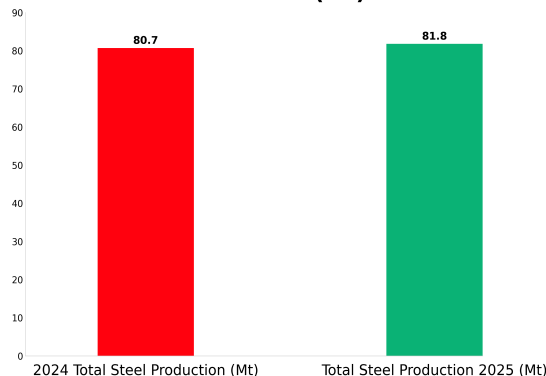
This type of misleading chart, published by an official government account with millions of followers, can affect how people judge whether a policy works. Viewers are likely to conclude that a dramatic industrial revival has taken place, when in reality the change is marginal. The intended consequence is political: to create the impression of a major achievement.

Total U.S. Steel Production (Mt) 2024 v. 2025



(a) Original White House chart (truncated y-axis)

Total U.S. Steel Production (Mt) 2024 v. 2025



(b) Corrected chart with y-axis starting at zero

Figure 1: Comparison of the original White House chart and a corrected version with a y-axis starting at 0. The original visualization exaggerates the magnitude of the increase by truncating the y-axis.

References

- [1] The White House, “American steel is BACK.,” X, January 2026. Available: <https://x.com/WhiteHouse/status/2017370992436249025>
- [2] Tufte, E. R. (1991), *The Visual Display of Quantitative Information*. (2nd edition), Graphics Press, USA, 1991, p. 57–69.

#4. Lead author: Christian Ainis, Calling BS on "America's Most Dangerous Dog Breeds", Statista (2018)".

(What is the claim?)

In 2018, Statista published an infographic titled *"America's Most Dangerous Dog Breeds"* based on data from DogsBite.org regarding fatal dog attacks in the United States between 2005 and 2017 [1]. The visualization shows that pit bulls account for 284 fatal attacks, or 66% of all fatalities, exceeding other dog breeds. Therefore, the visualization frames pit bulls as the most vicious dog breed.

(Why is it BS?)

This case presents an unfair comparison through a misleading visualization. The infographic reports the absolute counts of fatal attacks by breed, without normalizing for the total population of each breed in the US. A more common breed will naturally appear more often in raw fatality counts. Without relative values (e.g., percentages or a common denominator such as attacks per 100.000 dogs), the figure cannot claim what breeds are more dangerous. The article also ignores major uncertainties present in dog breed identification. Studies show that visual identification of pit bulls and mixed-breed dogs is unreliable, often grouping multiple bully-type breeds into the same "Pit Bull" category [2]. This inflates the reported numbers and creates unfair comparisons between a loosely defined "Pit Bull" group versus more clearly defined groups, such as German Shepherds and Labradors. Finally, the article treats breed as the primary explanation for fatal attacks, giving little attention to alternative hypotheses. Research suggests that owners of "vicious" dogs are more likely to engage in criminal or antisocial behavior themselves [3]. Owner behavior, improper training, abuse, and neglect of the dog may therefore explain part of the observed differences rather than the breed.

(Which tools have you used?)

The tools of showing unfair comparisons and generating alternative hypotheses were applied to evaluate the visualization. Specifically, the analysis examined whether comparisons between breeds were statistically meaningful without normalization by breed population sizes. The critique also considered owner behavior and breed misidentification that may explain the observed pattern without concluding that pit bulls are inherently more dangerous.

(What are the (un)intended consequences of spreading this BS/misleading information?)

Presenting raw fatality counts without context reinforces stigma and fear towards certain dog breeds. Such claims are often used to justify breed-specific legislation and restrictions despite uncertainty in the underlying data. Misleading visualizations may distract attention from serious predictors of aggressive dog behavior, such as irresponsible ownership, poor training, and neglect of the dog.

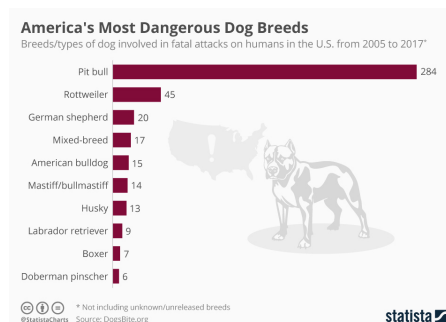


Figure 2: Statista infographic showing raw counts of fatal dog attacks by breed in the United States (2005-2017) without normalization by breed population.

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#4. Lead author: David Poda, Calling BS on "Violent Crime Plummet Across Major U.S. Cities", White House (2026)".

(What is the claim?)

The article that was published on the official White House website on the February 12, 2026 [1], claims that violent crime has significantly decreased across U.S. cities under President Donald J. Trump's political influences and commitments to restore law.

(Why is it BS?)

While the article is particularly highlighting declines such as 19% drop in homicides and other similar drops in other categories, the evidence that proves this causal relationship is not necessarily provided. The reported statistics that are shown are accurate, but they are presented in a misleading way and create an impression of causal link between the decline in crime and the current administration's policies. The article lacks a proper null model, as it does not establish what level of change would be expected without the influence of the current administration.

The main issue of the article is the claim suggested by it. By presenting the decline in crime alongside political statements, it is implied that the administration and current policies are the cause of the improvement. However, no such causal relationship is shown by providing concrete evidence for the case. The fact that crime declines happened at the same time as the new policies took effect does not prove that one caused the other without any other factors shown. This is a typical example of presenting correlation as a causation. It is also important to note that the article focuses on an extremely narrow time frame and shows only recent data, which further highlights the unfairness of the comparison.

(Which tools have you used?)

The concept of null model was applied to call out on this article, by asking whether the level of crime would be expected without the administration policies in place, based on already existing trends. Such baseline is not established in the article in any way, making it unclear if the current decline was caused by the new intervention. An alternative hypothesis would be asking whether other factors may explain the decline or not, which is something that is not considered in the article at all. According to another article Council on Criminal Justice [2], homicide rates began declining significantly already in years 2023–2024, well before the current policies have taken effect, which would suggest that the data is presenting a continuation of an ongoing trend rather than the result of new policies.

(What are the (un)intended consequences of spreading this BS/misleading information?)

These kinds of incorrect presentations can have a significant effect on how public perception is shaping. When official government sources present data in a misleading way, it can compromise public trust and make it harder for regular citizens to make sensible decisions about the effectiveness of political actions. This can in the long run lead to poor policy choices that do not actually address the cause of issue.

References

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